Venkata Lakshmi Parimala Pasupuleti

Full-Stack AI Engineer · Real Estate Domain · React · TypeScript · Python · Node.js · RAG · PDF Parsing · AWS

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PROFESSIONAL SUMMARY

Full-stack AI engineer with **8 years of real estate domain expertise** and 7+ years shipping production systems — exactly what Nura needs. At Entrata (multifamily & affordable housing SaaS, 100K+ MAU), built the data infrastructure behind **income verification, deal underwriting workflows, lease compliance, and lender-facing HUD/Tax Credit certifications** across **1.6M+ property records**. At Intuit, now ships **AI-powered features** using Anthropic Claude on Python/FastAPI with React/TypeScript frontends. Leasel is a production-grade system for **PDF parsing and RAG over unstructured real estate documents** — OCR ingestion, section-aware chunking, hybrid vector search, cited answers — achieving **93.6% precision** across 1,240 lease documents. AgentProbe routes prompts across 5 LLM providers with a 15-endpoint FastAPI backend. Stack: **React, TypeScript, Python, Node.js, AWS, RAG, ChromaDB**. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI & Data: LLMs (Anthropic Claude, OpenAI, Groq, Ollama), RAG Pipelines, PDF Parsing & OCR, Unstructured Data Extraction, Vector Search, ChromaDB, Pinecone, Hybrid Search (TF-IDF + Semantic), LLM-as-Judge Evaluation, LangGraph, LangChain, Prompt Engineering

Full-Stack: Python (9+ yrs), FastAPI, Flask, Node.js, React, Redux, TypeScript, JavaScript, REST APIs, GraphQL, PHP, Laravel, Celery, RabbitMQ

Cloud & Infrastructure: AWS (S3, EC2, RDS, Lambda, SQS, Glue, API Gateway, IAM), Docker, Kubernetes, GitHub Actions CI/CD, PostgreSQL, MySQL, Snowflake, Redis, MongoDB, Git, Linux

Real Estate & Fintech Domain: Multifamily & Affordable Housing, Income Verification, Lease Compliance, LIHTC/HCV/HUD Certifications, Lender-Facing Workflows, Property Management Systems, Financial Records at Scale

WORK EXPERIENCE

Intuit

Jun 2026 – Present

Software Engineer — AI-Powered Features · Full-Stack · Financial Analytics

- Build **AI-powered features** integrating Anthropic Claude into Python/FastAPI backend services and React/TypeScript frontends — financial data summaries, natural-language queries, and reporting; implement guardrails, LLM eval harnesses (LLM-as-judge + golden datasets), and CI/CD on AWS with Prometheus observability.
- Build end-to-end ML pipelines (scikit-learn, XGBoost) on Snowflake from data ingestion through prediction APIs; translate customer requirements into technical solutions shipped daily to production users.

Entrata — Multifamily & Affordable Housing SaaS

Jan 2017 – Jul 2024

Senior Software Engineer — Real Estate & Lender Data Infrastructure

- Built the full-stack systems behind **income verification, lease underwriting, and lender-facing HUD/Tax Credit certification workflows** across 1.6M+ property records — Python, PHP, Java, React, PostgreSQL; served as **primary technical contact for HUD and Tax Credit federal data source teams**, coordinating API integrations, debugging data issues, and driving go-lives end to end.
- Shipped 20+ features across **120+ Agile sprints**; async processing (RabbitMQ, idempotency, retries) for rent updates across thousands of tenant records; **optimized 200+ SQL queries (30% performance gain)** across 1M+ records; Docker microservices (50% faster execution); AWS S3 migration with RBAC for PHI — **100% compliance, zero incidents**; reviewed 300+ PRs.

PROJECTS

Leasel — RAG over Unstructured Real Estate Documents

PDF Parsing · RAG · Vector Search

Python · LangGraph (8-agent) · FastAPI · OCR · Hybrid RAG (TF-IDF + semantic) · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **PDF parsing pipeline** (OCR ingestion, section-aware chunking) + **hybrid vector search RAG** over 1,240 unstructured lease documents; 8-agent LangGraph workflow for compliance analysis; answers cited to exact source clauses; LLM-as-judge CI eval: **93.6% precision, 87.9% F1**. Direct analog to Nura's document intelligence layer for developers and lenders.

AgentProbe — LLM Router & Evaluation Toolkit (Open Source)

AI Features · Multi-Provider

Python · FastAPI · LangGraph · ChromaDB · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama — github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend routing prompts across 5 LLM providers; dynamic model routing; 5-metric eval engine (accuracy, hallucination, retrieval, completeness, compliance); ChromaDB vector search; CI golden-dataset regression harness. Ships to real users, zero external dependencies.

AI on FHIR — Full-Stack NLP-to-API with React + TypeScript

Full-Stack · React · TypeScript

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · REST APIs — github.com/venkatalakshmiparimala/AI_on_FHIR

- Full-stack AI app (React/TypeScript frontend + Python/Flask backend) translating EN/ES natural-language queries into live API calls via LLMs + BERT; HIPAA-style RBAC; Docker; cut manual lookup time ~60%.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · ML pipelines, LLM systems, statistical modeling, data governance, unstructured data

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017